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Roll No

BE-3001(CS)-CBGS

B.E., III Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Mathematics - III

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Find Fourier series expansion of the function

$$f(x) = |x|, -\pi < x < \pi.$$

2. Find Fourier sine transform of $e^{-|x|}$. Hence, evaluate

$$\int_0^{\infty} \frac{x \sin mx}{1+x^2} dx.$$

3. Find the Fourier transform of

$$f(x) = \begin{cases} 1-x^2, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases}$$

4. Evaluate $\int_0^{\infty} t e^{-3t} \sin t dt$

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5. Suppose a continuous random variable X has density function.

$$f(x) = \begin{cases} k(1-x^2) & \text{for } 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

- i) Find k
 - ii) Find $P(0.1 < x < 0.2)$
 - iii) $P(x > 0.5)$.
6. Find mean and variance of a continuous random variable X where density function is given by

$$f(x) = \begin{cases} \frac{1}{4}e^{-\frac{x}{4}}, & \text{for } x > 0 \\ 0, & \text{otherwise} \end{cases}$$

7. Find the Fourier series for $f(x) = \begin{cases} -\pi, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$

and deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$

8. a) Explain Poisson and Normal distribution.
b) Fit a second degree parabola to the following data:

x :	0	1	2	3	4
y :	1	1.8	1.3	2.5	6.3
